

# Advanced Paradigms in Chronic Kidney Disease Care: Integrating Clinical AI Agents, Decentralized Multi-Agent Systems, and Modern Pharmacological Combination Therapies

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## Executive Summary

The management of Chronic Kidney Disease (CKD) is undergoing a dual paradigm shift driven by advancements in artificial intelligence (AI) and novel pharmacological combination therapies. Historically, AI applications in nephrology have been limited to isolated predictive models that fail to integrate seamlessly into clinical workflows. To address this, a new paradigm of "Clinical AI Agents" has emerged—workflow-native, real-time systems designed to continuously perceive patient data, reason under clinical constraints, and support coordinated action. Concurrently, decentralized collaborative frameworks, such as the blockchain and explainable AI (XAI) system proposed by Puri et al. (2025), offer promising methods for multi-institutional model training without compromising patient privacy.

On the clinical front, modern guideline-directed medical therapies (GDMT)—including SGLT2 inhibitors, GLP-1 receptor agonists, and the non-steroidal mineralocorticoid receptor antagonist (ns-MRA) finerenone—have established new standards for reducing metabolic, cardiovascular, and renal risks. Effective CKD management requires precise, GFR-stratified antihypertensive regimens, structured alkali therapy, and rigorous clinical monitoring to prevent complications such as hyperkalemia and metabolic acidosis.

While these technological and pharmacological innovations hold immense potential, several challenges remain. The clinical efficacy of workflow-native AI agents remains largely conceptual, and decentralized frameworks have only been validated in simulated environments. Furthermore, the long-term safety profiles for emerging combination therapies are still undergoing clinical trial evaluation. This report synthesizes these cutting-edge developments, outlining their applications, advantages, current limitations, and future directions.

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## Introduction

Chronic Kidney Disease (CKD) is a progressive condition characterized by a gradual loss of kidney function, requiring highly coordinated, multidisciplinary care. Managing CKD involves controlling systemic hypertension, mitigating metabolic complications, and slowing disease progression through complex pharmacological interventions.

Traditionally, clinical decision support in nephrology has relied on static risk calculators and predictive AI models. While these models can predict adverse events—such as acute kidney injury (AKI) or progression to end-stage renal disease (ESRD)—they lack the capacity to translate predictions into real-time, coordinated clinical actions.

To bridge this gap, modern research focuses on workflow-native Clinical AI Agents and decentralized multi-agent systems. These technologies operate in tandem with advanced pharmacological strategies, including renin-angiotensin-aldosterone system (RAAS) blockade, alkali therapy, and novel combination regimens (e.g., SGLT2 inhibitors and finerenone). This report provides a comprehensive analysis of these intersecting domains, highlighting how intelligent systems and guideline-directed medical therapies are shaping the future of nephrology.

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# Key Concepts

## 1. Clinical AI Agents

Unlike traditional predictive AI, which operates as a one-way diagnostic tool, Clinical AI Agents are designed to be workflow-native. They feature: \* **Continuous Perception:** Ongoing monitoring of evolving patient data and clinical states. \* **Bidirectional Interaction:** Active engagement with both the clinical environment and healthcare providers. \* **Adaptive Reasoning:** The ability to make decisions under real-time clinical constraints and adapt to changing patient dynamics.

## 2. Decentralized Collaborative Frameworks

Proposed by Puri et al. (2025), this architecture combines blockchain technology with Explainable AI (XAI) to facilitate collaborative model training across multiple healthcare institutions. Key components include: \* **Privacy Preservation:** Utilizing blockchain to train predictive models collaboratively without exposing sensitive patient data. \* **Explainability:** Integrating SHAP (SHapley Additive exPlanations) to provide transparent, interpretable insights for clinical decision support. \* **Multi-Agent Optimization:** Deploying independent agents to optimize distinct therapeutic dimensions, such as medication, diet, physical activity, and mental health.

## 3. Guideline-Directed Medical Therapy (GDMT)

Modern pharmacological management of CKD relies on a cornerstone triad of therapies designed to provide additive cardiorenal protection: \* **SGLT2 Inhibitors (SGLT2i):** Lower glucose levels and reduce intraglomerular pressure. \* **GLP-1 Receptor Agonists (GLP-1RAs):** Provide cardiovascular and metabolic benefits. \* **Non-Steroidal Mineralocorticoid Receptor Antagonists (ns-MRAs):** Agents like finerenone that reduce inflammation and fibrotic activity.

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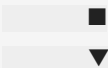
# Detailed Analysis

## Pharmacological Management & Combination Therapies

### Antihypertensive Therapy & RAAS Blockade

Hypertension control is critical in slowing CKD progression. Most patients require a multi-drug regimen to achieve target blood pressure.

Initial SBP  $\geq 150$  mmHg



Initiate Dual Therapy Immediately:

1. Anti-RAAS Agent (ACEI or ARB)

AND

2. Diuretic (Thiazide) OR Calcium Channel Blocker (CCB)

- **First-Line Selection:** Angiotensin-converting enzyme inhibitors (ACEIs) or angiotensin II receptor blockers (ARBs) are first-line agents.

- *Type 1 Diabetes*: ACEIs or ARBs.
- *Type 2 Diabetes*: ARBs or ACEIs.
- **Indications for RAAS Blockade**: Recommended for adults with diabetes and a urine albumin-to-creatinine ratio (uACR) of  $\geq 30$  mg/24h, or any adult with a uACR of  $\geq 300$  mg/24h.
- **Contraindications**: Dual therapy combining an ACEI with an ARB is strictly avoided due to elevated risks of hyperkalemia and acute kidney injury (AKI).

#### Line-by-Line Agent Selection Protocol

When blood pressure remains uncontrolled, subsequent agents are selected based on the patient's Glomerular Filtration Rate (GFR) and clinical presentation:

| Line of Therapy | Clinical Parameter                       | Recommended Agent(s)                                                                               |
|-----------------|------------------------------------------|----------------------------------------------------------------------------------------------------|
| 2nd & 3rd Line  | GFR $\geq 40$ mL/min/1.73 m <sup>2</sup> | Add Thiazide and/or Calcium Channel Blocker (CCB)                                                  |
| 2nd & 3rd Line  | GFR $< 40$ mL/min/1.73 m <sup>2</sup>    | Add Loop Diuretic (twice-daily bumetanide or furosemide, or once-daily torsemide) and/or CCB       |
| 4th Line        | Heart Rate $> 80$ bpm                    | Beta-blocker or Alpha/beta-blocker (e.g., labetalol)                                               |
| 4th Line        | Heart Rate $\leq 80$ bpm                 | Aldosterone Receptor Antagonist (ARA) (e.g., spironolactone, eplerenone) if proteinuria is present |

#### Specific Clinical Comorbidities

- **Coronary Artery Disease (CAD):** Beta-blocker, CCB, or alpha/beta-blocker.
- **Benign Prostatic Hyperplasia (BPH):** Alpha-1 blockers (e.g., prazosin, terazosin, doxazosin).
- **Thiazide-Resistant Hypertension:** Amiloride or ARA.

### Alkali (Base) Therapy

Metabolic acidosis is a common complication of progressive CKD. Sodium bicarbonate ( $\text{NaHCO}_3$ ), available in 325 mg and 650 mg tablets, is utilized to prevent protein wasting, decrease bone demineralization, and increase vitamin D synthesis.

- **Initiation Criteria:** Initiate therapy if serum  $\text{HCO}_3^- < 22$  mEq/L on two separate occasions at least 2 weeks apart.
- **Dosing Regimen:** Administer 23–46 mEq/d in 2 to 3 divided doses (equivalent to 0.5–2.0 mEq  $\text{HCO}_3^-/\text{kg}/\text{d}$ ) to maintain a target serum  $\text{HCO}_3^-$  level of 22–26 mEq/L.
- **Drug Interactions:** Do not co-administer sodium bicarbonate with calcium-based phosphorus binders or iron salts.
- **Nephrology Consultation:** Required if serum  $\text{HCO}_3^-$  remains  $< 22$  mEq/L despite therapy, or if the required  $\text{NaHCO}_3$  dose exceeds 46 mEq/d.

### Combination Therapies for Risk Reduction

Combining SGLT2 inhibitors, GLP-1RAs, and the ns-MRA finerenone provides additive organ protection. Notably, co-administration can mitigate specific drug-induced side effects; for instance, combining an SGLT2 inhibitor with finerenone significantly decreases the risk of treatment-induced hyperkalemia. Emerging combinations, such as adding aldosterone synthase inhibitors or endothelin receptor antagonists to standard RAAS and SGLT2 inhibition, have demonstrated additive antiproteinuric effects.

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## Applications

### 1. Workflow-Native Clinical AI Integration

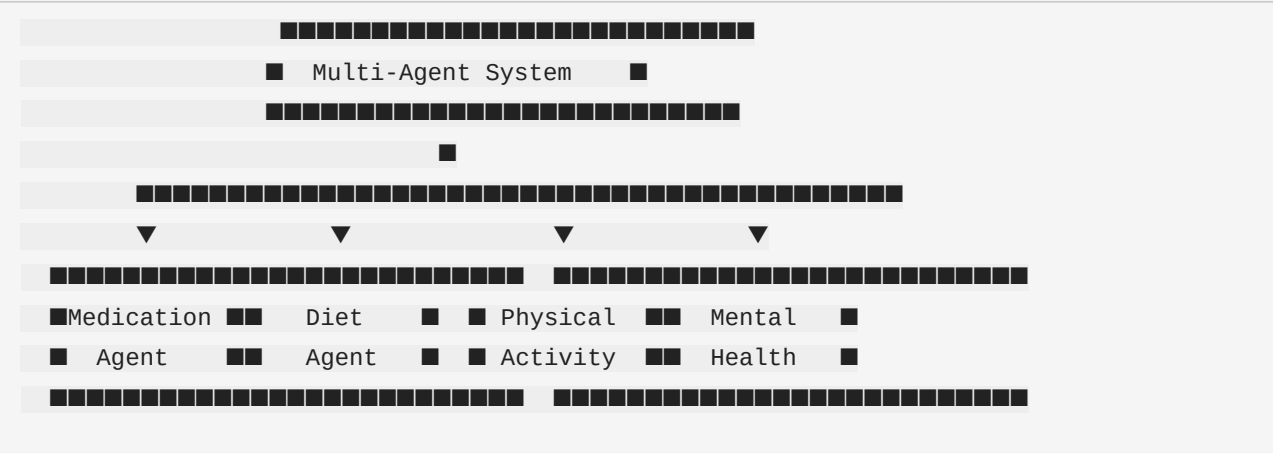
Clinical AI agents can be integrated directly into Electronic Health Records (EHR) to continuously monitor laboratory values (e.g., eGFR, serum potassium,  $\text{HCO}_3^-$ , and uACR). When a patient's parameters cross clinical thresholds, the agent can dynamically suggest medication adjustments, such as initiating alkali therapy or adjusting antihypertensive agents based on the GFR-stratified protocol.

## 2. Decentralized Collaborative CKD Detection

The framework proposed by Puri et al. (2025) allows geographically separated healthcare networks to collaboratively train predictive CKD models. By utilizing blockchain, institutions can share model weights rather than raw patient data, maintaining strict compliance with data privacy regulations.

## 3. Multi-Dimensional Patient Optimization

In a multi-agent system, specialized agents independently optimize distinct aspects of CKD care:



This ensures a holistic approach to patient wellness, addressing lifestyle modifications alongside pharmacological management.

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## Advantages

- **Actionable Clinical Support:** Transitioning from passive predictions to workflow-native AI agents ensures that critical clinical changes (e.g., a drop in eGFR or bicarbonate) trigger immediate, coordinated clinical recommendations.
- **Privacy-Preserving Collaboration:** The decentralized blockchain framework enables robust model training across diverse patient populations without exposing sensitive Protected Health Information (PHI).
- **Enhanced Clinical Trust:** Incorporating SHAP-based explainability ensures that AI-generated recommendations are transparent and interpretable, increasing clinician confidence in automated decision support.
- **Synergistic Pharmacological Effects:** Combining GDMT agents (such as SGLT2 inhibitors and finerenone) provides superior cardiorenal protection while actively mitigating adverse effects like hyperkalemia.

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## Challenges

### 1. Unproven Real-World AI Efficacy

The workflow-native Clinical AI Agent paradigm remains largely conceptual. Its real-world clinical utility, seamlessness of EHR integration, and long-term safety profiles have not yet been established in live clinical environments.

### 2. Simulation-Only Validation of Decentralized Frameworks

The decentralized collaborative framework proposed by Puri et al. (2025) has only been tested in simulated environments using public datasets (the CDC Chronic Disease and UCI Chronic Kidney Disease Risk Factor datasets). Its security, communication overhead, and performance under real-world clinical data variations remain unverified.

### 3. Complex Pharmacological Risks

- **Dual RAAS Blockade Risks:** Combining ACEIs and ARBs is contraindicated due to severe risks of hyperkalemia and AKI.
- **Drug-Drug Interactions:** Sodium bicarbonate cannot be co-administered with iron salts or calcium-based phosphorus binders, complicating medication scheduling for patients with advanced CKD.
- **Monitoring Burden:** Intensive combination therapies require frequent laboratory monitoring for hyperkalemia, metabolic acidosis, and rapid eGFR declines.

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## Future Scope

- **Clinical Validation of AI Frameworks:** Future research must transition Clinical AI Agents and decentralized frameworks from simulated environments to prospective clinical trials to evaluate their safety, usability, and impact on patient outcomes.
- **Ongoing Combination Therapy Trials:** Multiple clinical trials are planned or currently underway to evaluate the long-term safety, efficacy, and antiproteinuric benefits of emerging combinations, such as aldosterone synthase inhibitors and endothelin receptor antagonists.

- **Standardization of Evaluation Metrics:** Establishing standardized benchmarks for federated systems (e.g., communication overhead, convergence rounds) and reinforcement learning agents (e.g., reward scores, episode returns) will be critical for comparing diverse AI architectures in nephrology.

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## Conclusion

The integration of workflow-native Clinical AI Agents and decentralized collaborative frameworks represents a promising frontier in nephrology, offering the potential to translate complex patient data into timely, coordinated clinical actions. When paired with modern, guideline-directed combination therapies—such as SGLT2 inhibitors, finerenone, and structured GFR-stratified antihypertensive regimens—these technologies could significantly slow CKD progression and reduce cardiovascular complications. However, realizing this potential requires overcoming significant hurdles, including the clinical validation of conceptual AI models, managing complex drug interactions, and completing ongoing clinical trials to confirm the long-term safety of emerging pharmacological combinations.

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## References

### Study and Guideline References

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